

CUSTOMER CHURN ANALYSIS

Project Notes-1 Submission

Submitted By

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Review Parameters	Review points
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b) Need of the study/project	
c) Understanding business/social opportunity	
2)Data Report	2
a) Understanding how data was collected in terms of time, frequency and methodology	
b) Visual inspection of data (rows, columns, descriptive details)	
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b) Bivariate analysis (relationship between different variables , correlations)	
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d) Outlier treatment (if required)	
e) Variable transformation (if applicable)	
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INTRODUCTION TO BUSINESS PROBLEM

Problem Statement:

An E Commerce company or DTH provider is facing a lot of competition in the current market and it has become a challenge to retain the existing customers in the current situation. The company is losing its customers as the customers have stopped using the services. This in turn is impacting the business. Hence, the company wants to develop a model through which they can do churn prediction of the accounts and provide segmented offers to the potential churners. In this company, account churn is a major thing because 1 account can have multiple customers. Hence by losing one account the company might be losing more than one customer.

We have a data extract containing the customer churn details. This helps us to analyze various features which might have attributed to customer churn.

Need of Study / Project:

Aim: To build a prediction model using Machine Learning method that could help predict the customer account churn and this in turn will help to identify the potential churners at the right time and provide segmented offers in order to retain them. The main aim is to prevent the customer churn by analyzing the data extract which has historical data of customer churn details. A churn model is **a mathematical representation of how churn impacts the business**. Churn calculations are built on existing data (the number of customers who left the service during a given time period). A predictive churn model extrapolates on the existing data to show future potential churn rates.

Understanding Business/ Social Opportunity

Every business deals with churn, and more often than not, it's much easier and less costly to keep an existing customer than it is to gain a new customer. It is also much easier to save a customer before they leave than it is to convince the customer to come back. Customer loyalty is something all brands strive for - understanding and preventing churn is critical to achieving this.

Data Report:

Understanding of Attributes:

Account ID:	It is the unique account identifier of the customer account
Churn:	Whether the customer account churn or not. 0 means No and 1 means Yes. It is also the Target variable
Tenure:	Tenure of account. We can assume the tenure to be in months
City Tier:	City Tier of the primary customer of the account. Here we have three levels of city tier
CC_Contacted_L12m	How many times all the customers of the account has contacted customer care in last 12months
Payment	Preferred Payment mode of the customers in the account. We have five type of Payment mode- Debit Card, Credit Card, E Wallet, Cash on Delivery and UPI
Gender	Gender of the primary customer of the account. We have Male and Female as genders
Service_ Score	Satisfaction score given by customers of the account on service provided by company
Account_ user_ count	Number of customers tagged with this account
account_ segment	Account segmentation on the basis of spend. We have five segments of customers Super, Super Plus, Regular, Regular Plus and HNI
CC_ Agent_ Score	Satisfaction score given by customers of the account on customer care service provided by company
Marital_ Status	Marital status of the primary customer of the account
rev_ per_ month	Monthly average revenue generated by account in last 12 months
Complain_l12m	Any complaints has been raised by account in last 12 months
rev_ growth_ yoy	revenue growth percentage of the account (last 12 months vs last 24 to 13 month)
coupon_ used_ l12m	How many times customers have used coupons to do the payment in last 12 months
Day_ Since_ CC_ connect	Number of days since no customers in the account has contacted the customer care
cashback_ l12m	Monthly average cashback generated by account in last 12 months
Login_ device	Preferred login device of the customers in the account

Importing the Libraries and Read the dataset into the Jupyter Notebook. Used Head and Tail function to look at the dataset.

The dataset has 11,260 rows and 19 columns. Here Churn column is the Target Variable

Account ID column is not a very important column from modeling purpose and can be dropped

On analyzing the Churn variable we see that more than 80% of values are 0 and around 16% with 1 value. Similarly the Cashback column has high no of missing values. The dataset is imbalanced type and will later need to be treated for model building and deployment

```
0    0.831616
1    0.168384
```

Descriptive Analysis

Get the info on the datatype of the columns as below:

#	Column	Non-Null	Count	Dtype
0	AccountID	11260	non-null	int64
1	Churn	11260	non-null	int64
2	Tenure	11158	non-null	object
3	City_Tier	11148	non-null	float64
4	CC_Contacted_LY	11158	non-null	float64
5	Payment	11151	non-null	object
6	Gender	11152	non-null	object
7	Service_Score	11162	non-null	float64
8	Account_user_count	11148	non-null	object
9	account_segment	11163	non-null	object
10	CC_Agent_Score	11144	non-null	float64
11	Marital Status	11048	non-null	object
12	rev_per_month	11158	non-null	object
13	Complain_ly	10903	non-null	float64
14	rev_growth_yoy	11260	non-null	object
15	coupon_used_for_payment	11260	non-null	object
16	Day_Since_CC_connect	10903	non-null	object
17	cashback	10789	non-null	object
18	Login_device	11039	non-null	object

dtypes: float64 (5), int64(2), object(12)

memory usage: 1.6+ MB

Observations:

1. There are 5 float type columns, 2 int64 type column and 12 object type columns
2. we can see that variables like 'Tenure', Account_user_count, rev_per_month, rev_growth_yoy, days_since_cc_connect and cashback are numerical but have been categorized as object above. We will need to evaluate the variables deeper and process them. Also we see that most of the attributes have missing values. further use isna function to get the details The feature Complain_Ly is float type but it should be ideally object type as it says if the customer has raised any complaints, yes or no.

Checking for Missing Values:

```
AccountID          0
Churn              0
Tenure            102
City_Tier         112
CC_Contacted_LY   102
Payment           109
Gender            108
Service_Score     98
Account_user_count 112
account_segment    97
CC_Agent_Score    116
Marital_Status    212
rev_per_month     102
Complain_ly       357
rev_growth_yoy     0
coupon_used_for_payment 0
Day_Since_CC_connect 357
cashback          471
Login_device      221
dtype: int64
```

Observations: Though we see that most of the columns have missing values the no of missing value is not much. Ideally we could have dropped the missing values but the dataset is not very big and dropping the missing values will decrease the dataset and hence we impute the missing values.

Describe Function

	count	mean	std	min	25%	50%	75%	max
AccountID	11260.0	25629.500000	3250.626350	20000.0	22814.75	25629.5	28444.25	31259.0
Churn	11260.0	0.168384	0.374223	0.0	0.00	0.0	0.00	1.0
City_Tier	11148.0	1.653929	0.915015	1.0	1.00	1.0	3.00	3.0
CC_Contacted_LY	11158.0	17.867091	8.853269	4.0	11.00	16.0	23.00	132.0
Service_Score	11162.0	2.902526	0.725584	0.0	2.00	3.0	3.00	5.0
CC_Agent_Score	11144.0	3.066493	1.379772	1.0	2.00	3.0	4.00	5.0
Complain_ly	10903.0	0.285334	0.451594	0.0	0.00	0.0	1.00	1.0

Observations:

1. the count for most of the numerical features is less than 11260, which means they have missing values
2. the churn value has imbalanced data as can be seen from the output above. More than 75% have 0 values
3. About 75% of the customers have rated the services 3 and the minimum score is 0. This means that customers are not happy with services

4. Mean score for service score and agent score is 3 which is less
5. Max customers are from tier 3 city
6. Max no of times since Customer Care was contacted in last 12 months is 132 and mean is 18 times which is average of 11 and 1 per month

Converting Column Complain_Ly into object datatype

```
0.0    0.714666
1.0    0.285334
```

Checking for Duplicate Rows

There are no duplicate rows

Percentage of Unique Count in the Columns

No of unique values in each column are as follows: (In Descending order)

```
Cashback: 5693: 50.56%
rev_per_month: 59: 0.52%
CC_Contacted_LY: 44: 0.39%
Tenure: 38: 0.34%
Day_Since_CC_connect: 24: 0.21%
rev_growth_yoy: 20: 0.18%
coupon_used_for_payment: 20: 0.18%
Account_user_count: 7: 0.06%
account_segment: 7: 0.06%
Service_Score: 6: 0.05%
Payment: 5: 0.04%
CC_Agent_Score: 5: 0.04%
Gender: 4: 0.04%
City_Tier: 3: 0.03%
Marital_Status: 3: 0.03%
Login_device: 3: 0.03%
Churn: 2: 0.02%
Complain_Ly: 2: 0.02%
```

Observations

Though we see most of the columns having unique counts the percentage is not much. Cashback has highest percentage of 51% of unique count. Further we will need to inspect the unique count and type of special characters or bad values

Next we do the Unique Value Count and have following Observation

Observation:

1. Columns like Tenure, Account_user_count, rev_per_month, coupon_used_for_payment, Days_since_CC_connect. cashback and rev_growth_yoy have special characters
2. Columns like account_segment, gender, login_device which are categorical columns also have data errors and need to be replaced

We have replaced the special characters in the numerical columns which show object type datatype with null values. This also converts the datatype from object type to float type

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11260 entries, 0 to 11259
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Churn                                11260 non-null  int64
1   Tenure                              11042 non-null  float64
2   City_Tier                           11148 non-null  float64
3   CC_Contacted_LY                     11158 non-null  float64
4   Payment                             11151 non-null  object
5   Gender                              11152 non-null  object
6   Service_Score                       11162 non-null  float64
7   Account_user_count                  10816 non-null  float64
8   account_segment                     11163 non-null  object
9   CC_Agent_Score                      11144 non-null  float64
10  Marital_Status                      11048 non-null  object
11  rev_per_month                       10469 non-null  float64
12  Complain_ly                         10903 non-null  object
13  rev_growth_yoy                      11257 non-null  float64
14  coupon_used_for_payment              11257 non-null  float64
15  Day_Since_CC_connect                10902 non-null  float64
16  cashback                            10787 non-null  float64
17  Login_device                        11039 non-null  object
dtypes: float64(11), int64(1), object(6)
memory usage: 1.5+ MB
```

We observe that most of the numerical columns which were shown object type have been reverted back to float type. These columns are Tenure, Account_user_count, rev_per_month, coupon_used_for_payment, Days_since_CC_connect, cashback and rev_growth_yoy

Unique Value of the Object Type Data Columns

```
Payment
Debit Card      4587
Credit Card    3511
E wallet        1217
Cash on Delivery 1014
UPI              822
Name: Payment, dtype: int64
```

```
Gender
Male      6328
Female    4178
M          376
F          270
Name: Gender, dtype: int64
```

```
account_segment
Super      4062
Regular Plus 3862
HNI        1639
Super Plus  771
```

```
Regular          520
Regular +        262
Super +          47
Name: account_segment, dtype: int64
```

```
Marital_Status
Married          5860
Single           3520
Divorced          1668
Name: Marital_Status, dtype: int64
```

```
Complain_ly
0.0           7792
1.0           3111
Name: Complain_ly, dtype: int64
```

```
Login_device
Mobile           7482
Computer         3018
&&&&             539
Name: Login_device, dtype: int64
```

We observe the presence of Bad Data in Account_Segment, Gender, Complain_Ly and Login Device

Use Replace function to remove the anomalies from these columns

Impute the Missing Values with mode for Categorical Columns like Gender, Payment, Account_Segment, Complain_Ly, Login_Device

Similarly impute the missing values for numerical columns with median values. These columns are Tenure', 'City_Tier', 'CC_Contacted_LY', 'Service_Score', 'Account_user_count', 'CC_Agent_Score', rev_per_month, 'rev_growth_yoy', 'coupon_used_for_payment', 'Day_Since_CC_connect', cashback

Checking the Missing Values

```
Churn           0
Tenure          0
City_Tier       0
CC_Contacted_LY 0
Payment         0
Gender          0
Service_Score   0
Account_user_count 0
account_segment 0
CC_Agent_Score  0
Marital_Status  0
rev_per_month   0
Complain_ly     0
rev_growth_yoy  0
coupon_used_for_payment 0
Day_Since_CC_connect 0
cashback        0
```

Login_device

0

We observe that the Missing Values have been imputed well and the output shows 0 null values.

➡ Using Describe function to have a relook at column stats

	count	mean	std	min	25%	50%	75%	max
Churn	11260.0	0.168384	0.374223	0.0	0.00	0.00	0.00	1.0
Tenure	11260.0	10.985879	12.757534	0.0	2.00	9.00	16.00	99.0
City_Tier	11260.0	1.647425	0.912763	1.0	1.00	1.00	3.00	3.0
CC_Contacted_LY	11260.0	17.850178	8.814851	4.0	11.00	16.00	23.00	132.0
Service_Score	11260.0	2.903375	0.722476	0.0	2.00	3.00	3.00	5.0
Account_user_count	11260.0	3.704973	1.004383	1.0	3.00	4.00	4.00	6.0
CC_Agent_Score	11260.0	3.065808	1.372663	1.0	2.00	3.00	4.00	5.0
rev_per_month	11260.0	6.266874	11.488990	1.0	3.00	5.00	7.00	140.0
rev_growth_yoy	11260.0	16.193073	3.757271	4.0	13.00	15.00	19.00	28.0
coupon_used_for_payment	11260.0	1.790409	1.969331	0.0	1.00	1.00	2.00	16.0
Day_Since_CC_connect	11260.0	4.581261	3.649643	0.0	2.00	3.00	7.00	47.0
cashback	11260.0	194.934722	174.977865	0.0	147.89	165.25	197.31	1997.0

- The descriptive statistical analysis shows that Tenure has mean period as ~11 whereas maximum is 99.
- Minimum Tenure is 0 which seems to be an anomaly
- Maximum customer account is from Tier 3 city and median with tier 1 city
- No of times customers of an account contacted the customer care in last 12 month is maximum with 132 and latest 4
- Maximum score given by customers for service is 5 and the median is 3
- Maximum no of customers tagged in an account is 6 and median is 4 median rev per month from an account is 5

Data Visualization

Univariate Analysis:

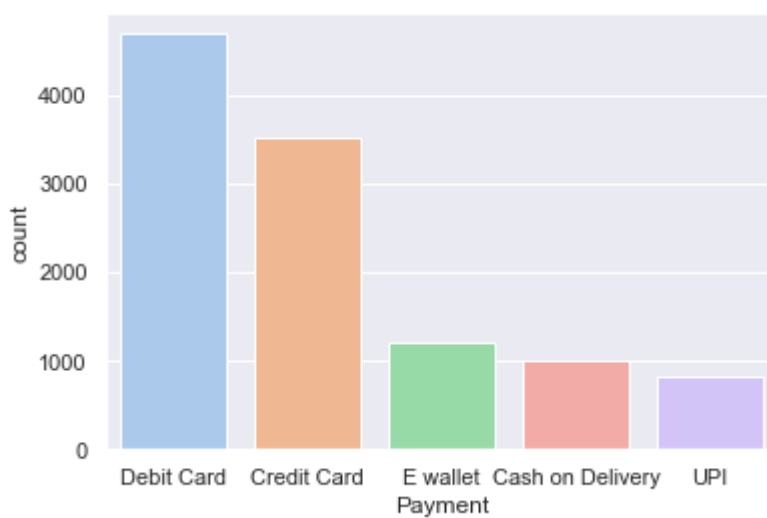
1. Count plot of Payment Type

No of unique values in **Payment** are: 5.

The distribution is as follows:

Payment

Debit Card	4696
Credit Card	3511
E wallet	1217
Cash on Delivery	1014
UPI	822



Univariate Analysis of Gender

2. No of unique values in **Gender** are: 2.

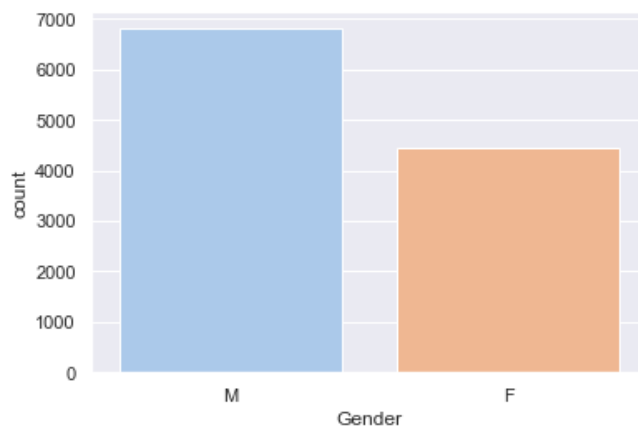
The distribution is as follows:

Gender

M 6812

F 4448

Name: Gender, dtype: int64



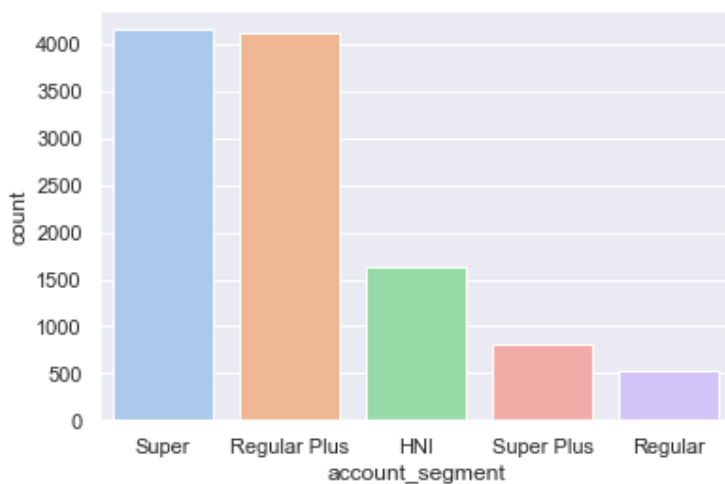
Univariate Analysis of account_segment

3. No of unique values in **account_segment** are: 5.
The distribution is as follows:

account_segment

Super	4159
Regular Plus	4124
HNI	1639
Super Plus	818
Regular	520

Name: account_segment, dtype: int64



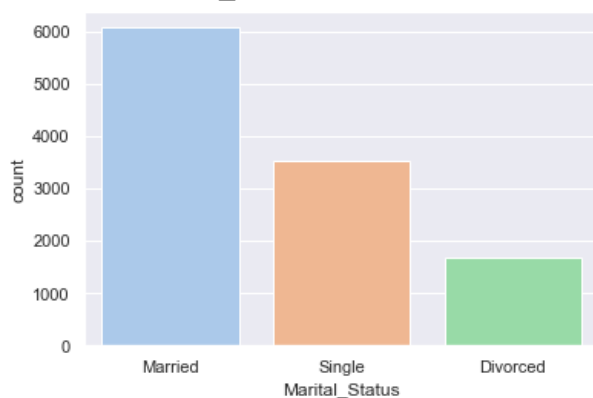
Univariate Analysis of Marital_Status

4. No of unique values in **Marital_Status** are: 3.
The distribution is as follows:

Marital_Status

Married	6072
Single	3520
Divorced	1668

Name: Marital_Status, dtype: int64



Univariate Analysis of Complain_ly

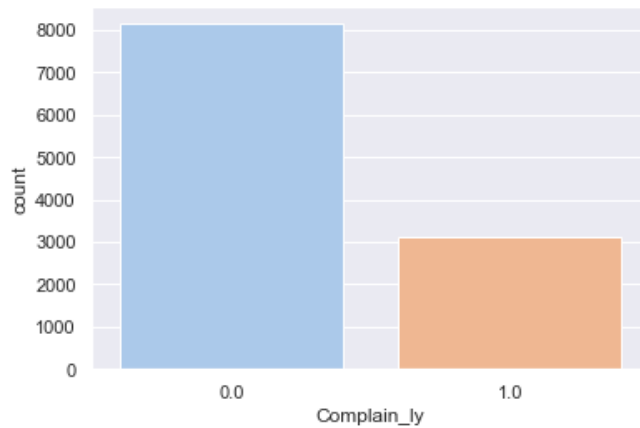
5. No of unique values in **Complain_ly** are: 2.
The distribution is as follows:

Complain_ly

0.0 8149

1.0 3111

Name: Complain_ly, dtype: int64



Univariate Analysis of Login_device

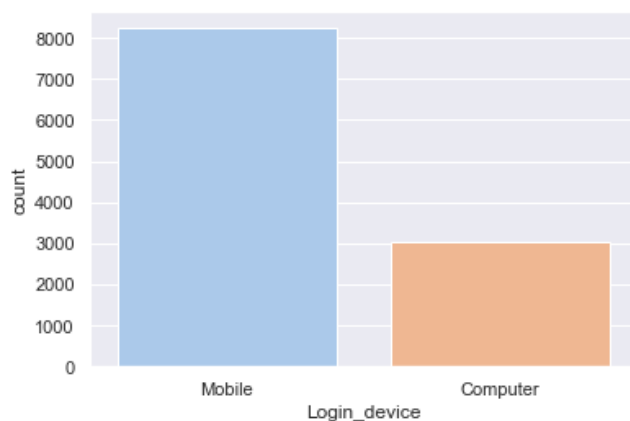
6. No of unique values in **Login_device** are: 2.
The distribution is as follows:

Login_device

Mobile 8242

Computer 3018

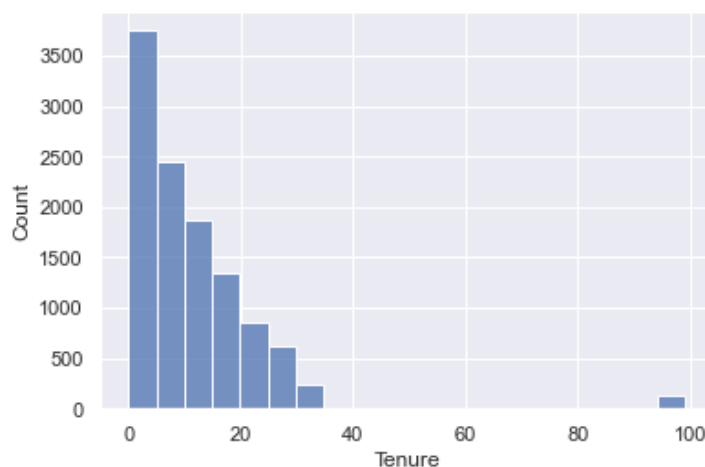
Name: Login_device, dtype: int64



Observations:

- The Payment made by Debit Card is highest at 4696 times followed by Credit Card at around 3511 times
- Gender wise Male customer count is higher at 6812 and Female at 4448
- Account_ Segment wise Super type has highest customers followed by Regular Plus and HNI
- Marital Status wise Married type are maximum amongst the customers
- Complaints made last Year has more customers with 0 or No complaints with ratio of ~ 80:20 of No: Yes
- The Login Device have more customers buying through mobile app than Laptop

Histogram of Tenure

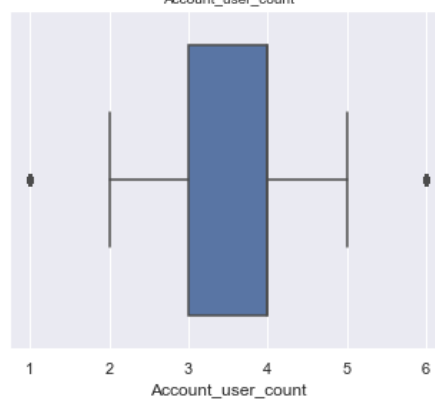
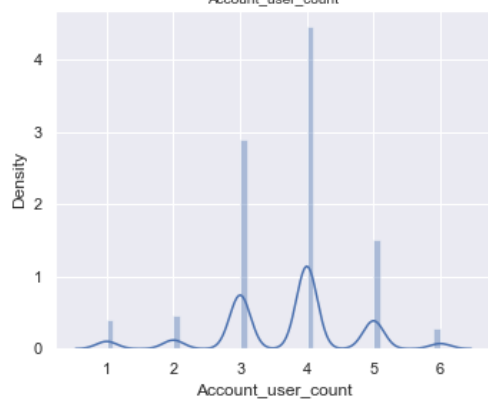
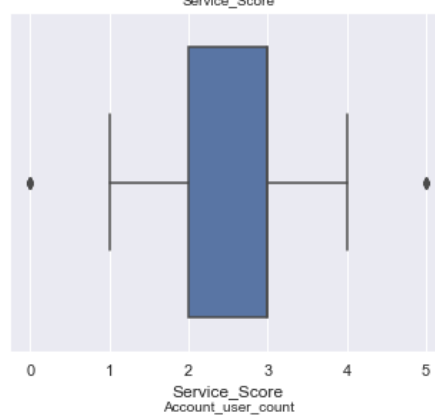
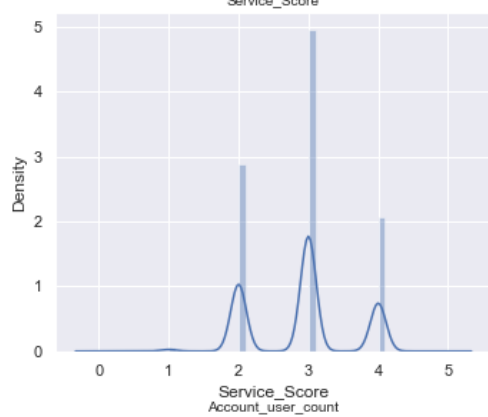
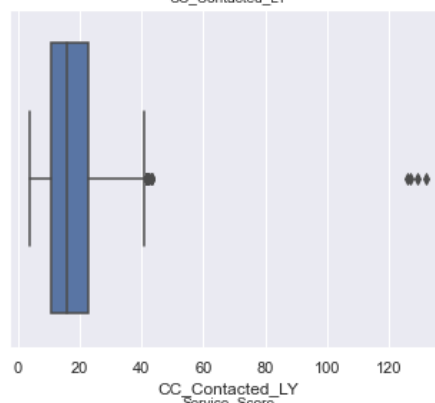
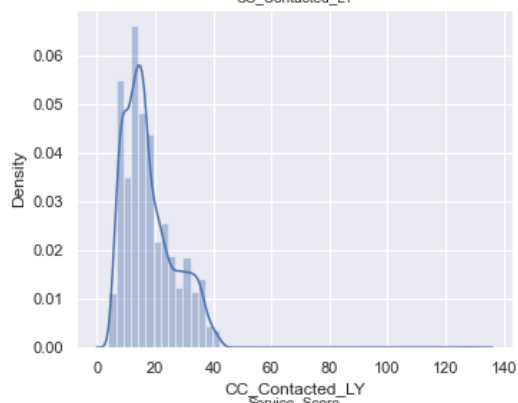
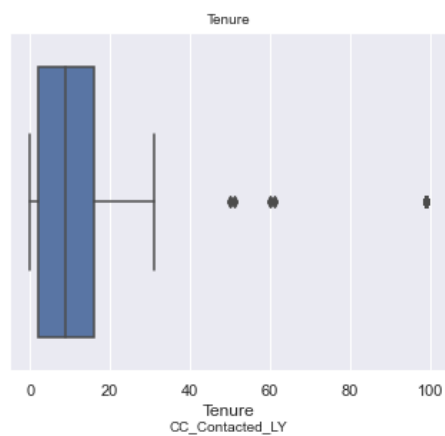
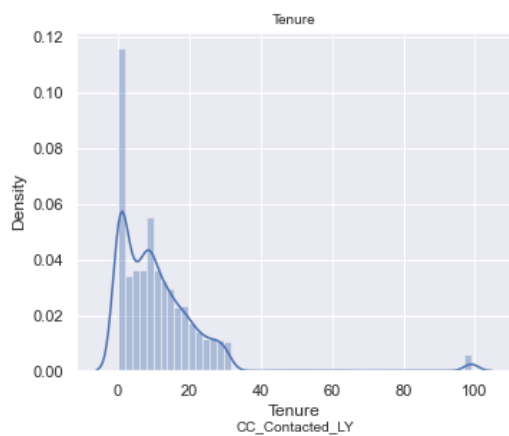


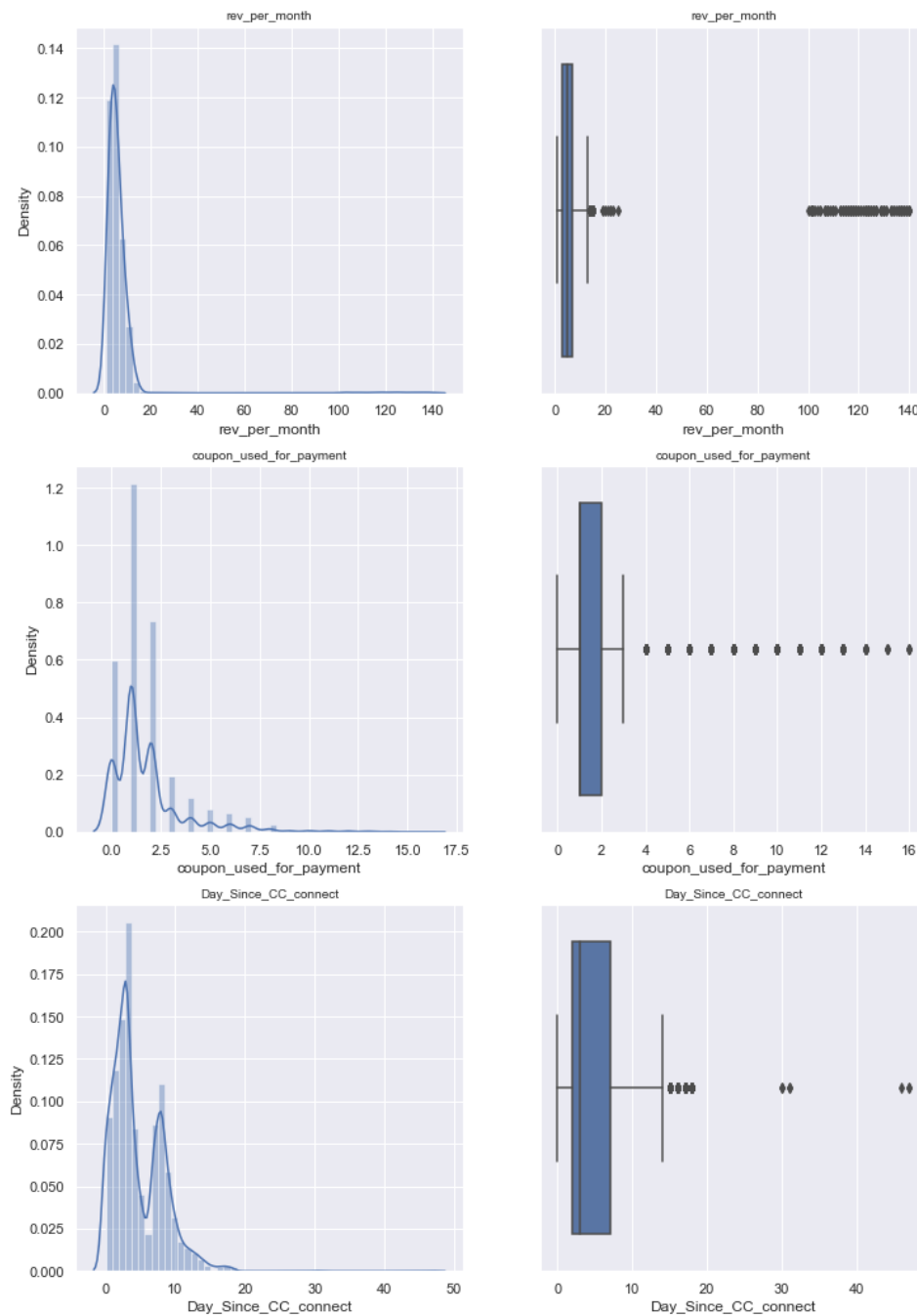
The histogram for the variable Tenure shows the Tenure to be spread from 0 to 100 months. Most of the distribution is cluttered between 0 and 20 months and secondarily between 0 and 40 months. The distribution seems right skewed few nos. can be seen between 95 and 100 months.

Comparative Visualization of Boxplot Vs Histogram Distribution for Numerical Columns

Observation:

1. The distribution for Tenure is Rightly skewed with distribution spread from 0 to 100. The boxplot has outliers with median tenure at 11 months.
2. Customer Care Contacted Last Year is Rightly skewed with spread from 0 to 140. The Maximum spread is between 0 and 45. The boxplot has median value at 20 and there are outliers present.
3. The distribution shows three peaks between 2 and 4 service scores. The boxplot for service score has Outliers and the median value is at 2.5.
4. Account User count has distribution spread between 1 to 6 and the peak is at 4 customers per account. The boxplot shows the presence of outliers and the median value is at 3.5.

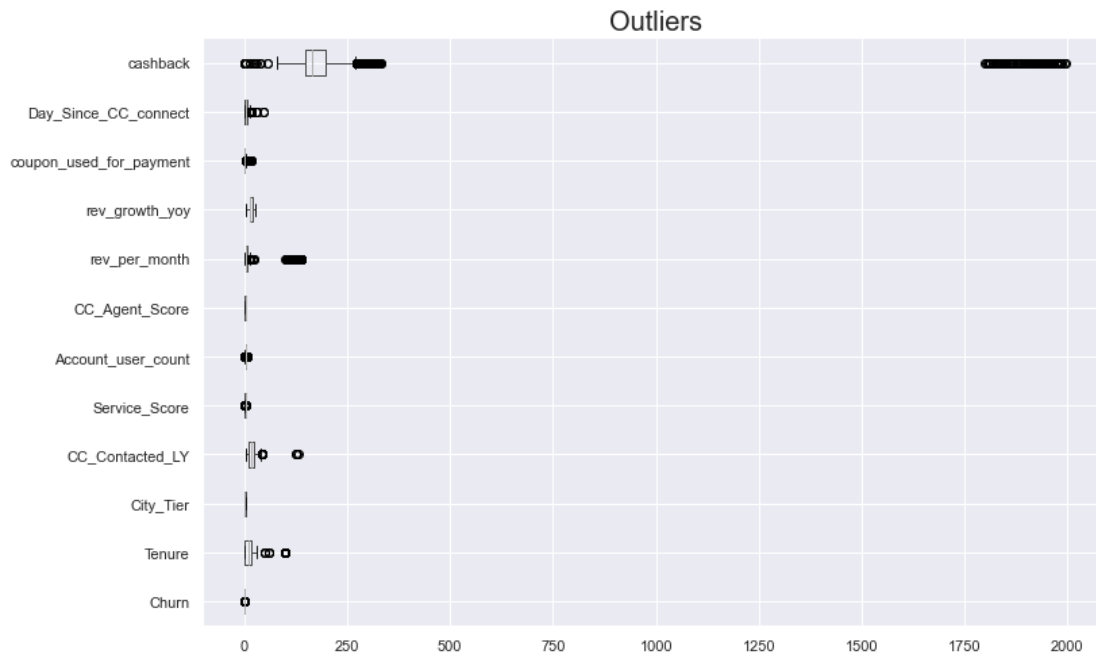




Observations:

1. Rev per month has spread from 0 to 140 with max revenue between 0 and 20. The distribution is rightly skewed. The boxplot has outliers and the median is at 10
2. Coupon used for payment has a very wavy distribution with spread from 0 to 17.5. The boxplot has outliers and the median is at 2
3. Days Since CC connect has two peak distribution spreading from 0 to 50. The boxplot has outliers and median is 0 and 1

Checking for Outliers using Boxplot



Observations:

1. Columns Churn, Coupon_used_for_payment and Cashback have max outliers
2. City Tier, Customer Care Agent Score, Revenue Growth YOY have no outliers
3. Other columns have minimal outliers

Bivariate Analysis

Pairplot for Numerical Columns

Based on the regression lines we can conclude the following correlation between the variables

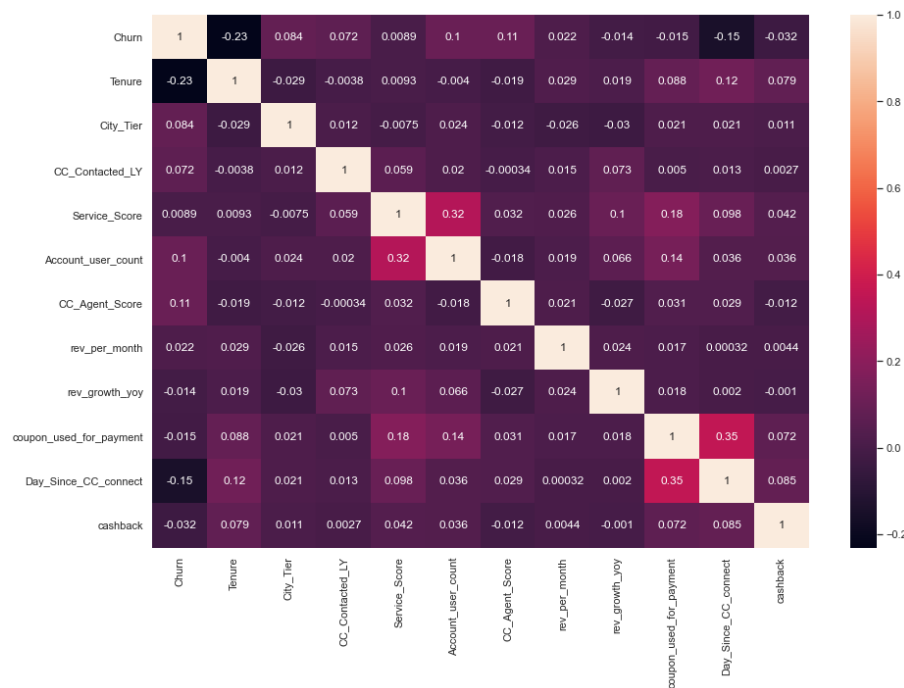
1. Churn and Tenure
2. Churn and Account_user_count
3. Churn and CC_Agent Score
4. Churn and Service_Score
5. Churn and Day_since_CC_connect
6. Churn and Cashback



Heat map to show correlation between variables

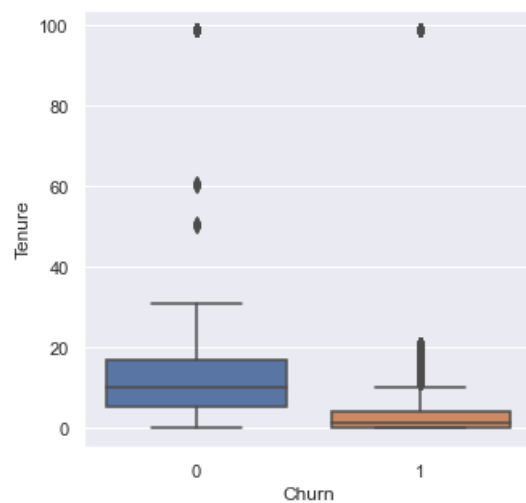
Insights:

1. Most of the variables have very less correlation between each other
2. The only variable which has some significant correlation ~ 0.35 is between Days since last connect and coupon used for payment. But since the correlation is less than 0.5 we may not need to drop the variables.
3. There is minimal correlation (negative) between Days since last connect and Churn and between Churn and Tenure
4. Other Variables which have comparatively better correlation are between CC Agent Score and Churn and secondly Tenure and days since last connect. We have also good correlation between coupon used for payment and service score and between coupon used for payment and account user count.

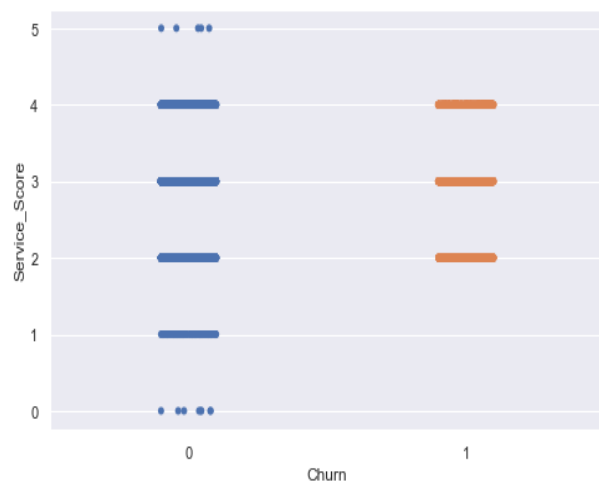


Boxplot showing Bivariate Relation

1. Churn and Tenure



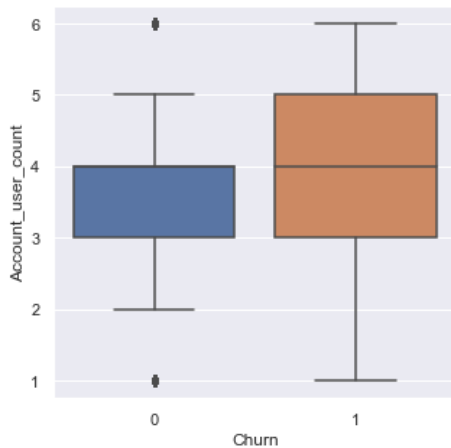
2. Churn and Service Score



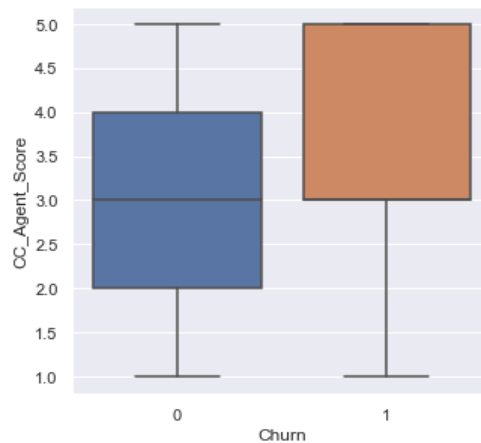
Here we have compared Churn against Tenure. Customers who have churned have had a very less tenure(0 to 1 month)which is less than median of customers who have not churned at 10 months

Accounts that have churned service score from 2 to 4. The major difference between account that have churned Vs against not churned is that service score is as low as 0 to as high as 5 for accounts that have not churned. Service Rating may be one of the reasons for churning

3 Churn and Account User Count



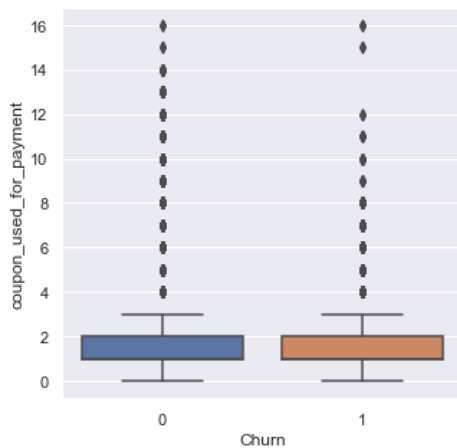
4 Churn and CC Agent Score



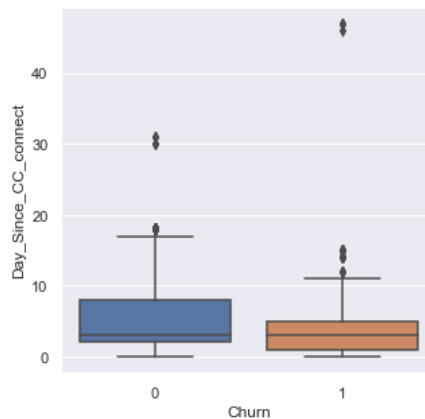
The median no of users per account for accounts that churned is at 4 which is higher than account user count that have not churned. This means that accounts that have churned have higher number of users and hence caused more loss

The median for Customer Care Agent Score given by users of account that have churned is higher at 4 as compared to median value of 3 given by account user that have not churned. This also means that customer care agent score is not one of the major issues for churning

5 Churn and Coupon used for Payment



6 Churn and Days Since CC Connect



Coupon Used for Payment does not show much variance between accounts that have churned and the ones that have not churned

The column Days since CC connect has equivalent median value for accounts that churned and not churned. The latest no of days connected by the user is 0 to 5 days. This means that even this variable does not hold much importance in terms of churning reasons

Bivariate Analysis using Crosstab

1. Churn & Complain_ Ly

Churn	0	1	All
Complain_ly			
0.0	0.643073	0.080639	0.723712
1.0	0.188544	0.087744	0.276288
All	0.831616	0.168384	1.000000

2. Churn & Account Segment

Churn	0	1	All
account_segment			
HNI	0.122913	0.022647	0.145560
Regular	0.042629	0.003552	0.046181
Regular Plus	0.266163	0.100089	0.366252
Super	0.330817	0.038544	0.369361
Super Plus	0.069094	0.003552	0.072647
All	0.831616	0.168384	1.000000

3. Churn & Payment

Churn	0	1	All
Payment			
Cash on Delivery	0.067496	0.022558	0.090053
Credit Card	0.267496	0.044316	0.311812
Debit Card	0.352753	0.064298	0.417052
E wallet	0.083570	0.024512	0.108082
UPI	0.060302	0.012700	0.073002
All	0.831616	0.168384	1.000000

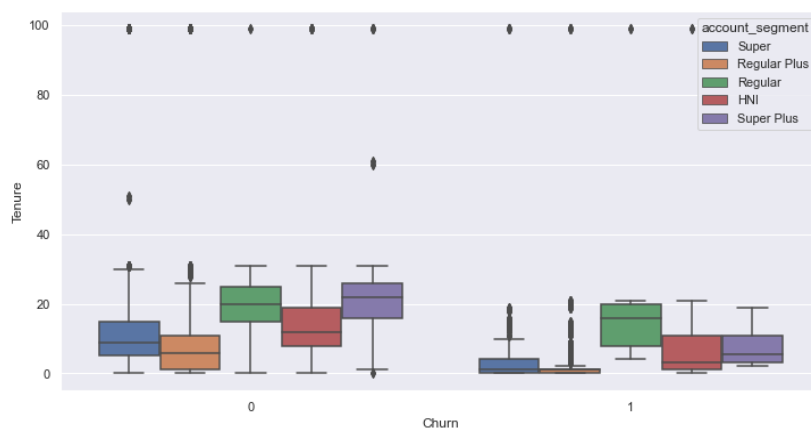
4. Churn & Tenure

Churn	0	1	All
Tenure			
0.0	562	669	1231
1.0	681	670	1351
2.0	322	32	354
3.0	367	43	410
4.0	360	43	403

5.0	373	30	403
6.0	348	15	363
7.0	418	32	450
8.0	490	29	519
9.0	645	69	714
10.0	396	27	423
11.0	368	20	388
12.0	347	13	360
13.0	334	25	359
14.0	320	25	345
15.0	291	20	311
16.0	272	19	291
17.0	209	6	215
18.0	242	11	253
19.0	249	24	273
20.0	187	30	217
21.0	150	20	170
22.0	151	0	151
23.0	169	0	169
24.0	147	0	147
25.0	114	0	114
26.0	122	0	122
27.0	131	0	131
28.0	137	0	137
29.0	114	0	114
30.0	137	0	137
31.0	96	0	96
50.0	2	0	2
51.0	2	0	2
60.0	2	0	2
61.0	2	0	2
99.0	107	24	131
All	9364	1896	11260

Multivariate Analysis

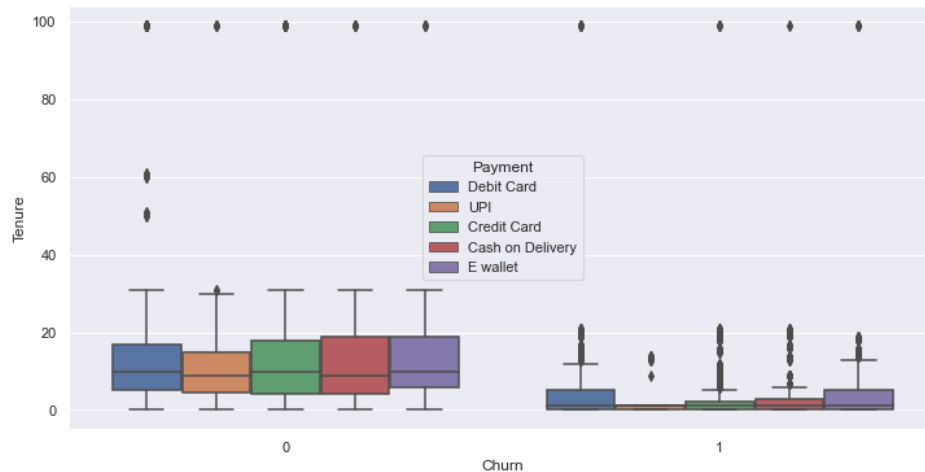
Factor Plot between Churn, Tenure and Account Segment



Insights:

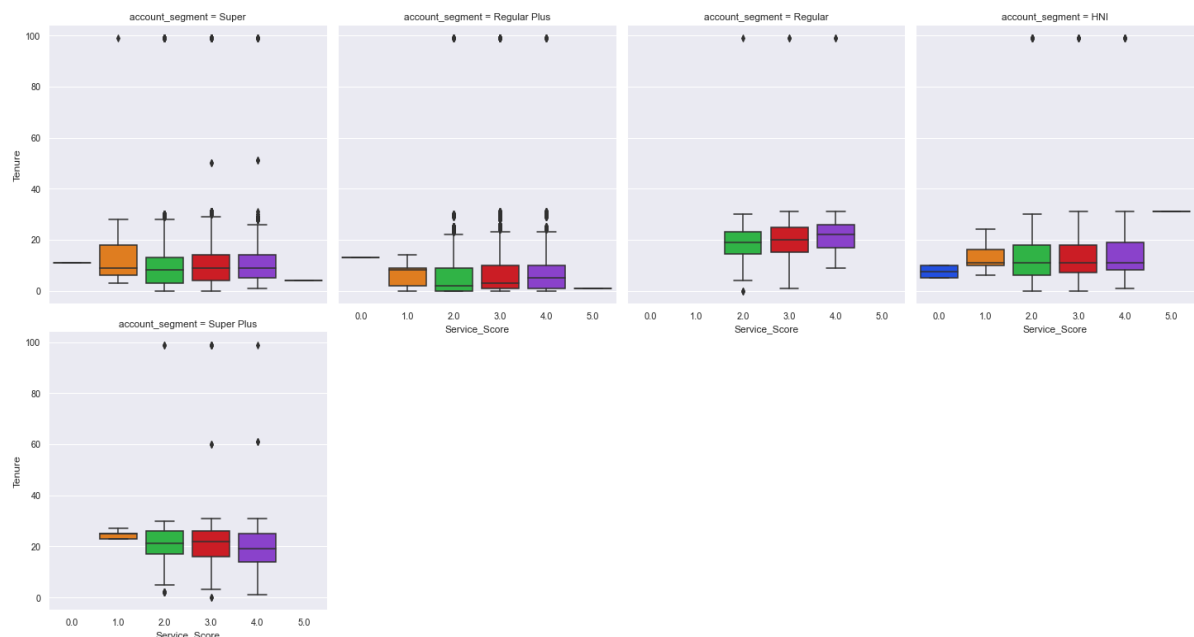
Comparison between Churn, Tenure and account segment shows maximum churn for Regular Category with median around 18 months. Other users have HNI and Super Plus Category. Outliers are present. Users who have not churned have highest median value at 22 for Super Plus category.

2. Factor plot between Churn, Tenure and Payment



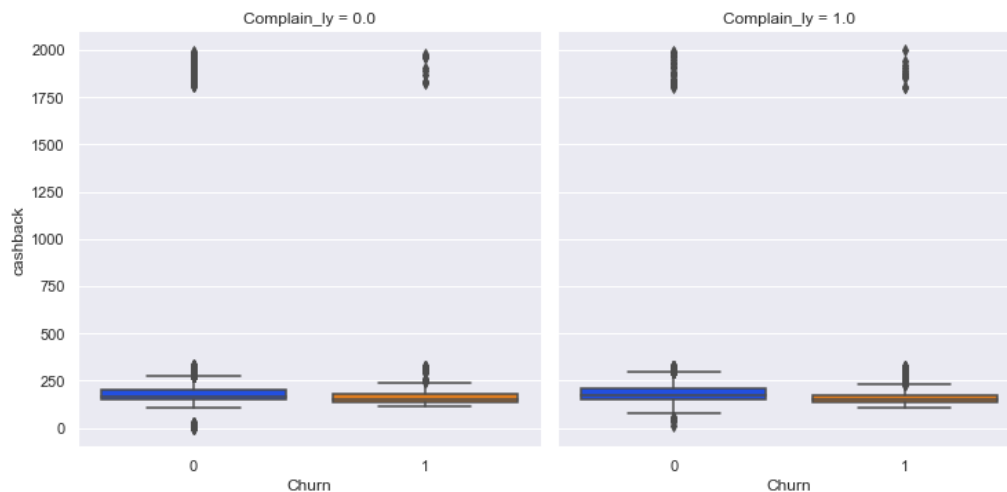
Comparison between Churn, Tenure and Payment shows maximum users with Debit Card and E Wallet type payment who have churned with tenure close to 0 and 1 month. Users who have not churned have all five types of users with max median value for Credit Card and Debit Card users with tenure ranging from 0 to 20 months

3. Factor plot for Service Score, Tenure and Account Segment



1. The Highest median for tenure is with Regular Category with median tenure at 21 at service score of 4.
2. Most of the service score ranges between 1 and 4 with outliers present for service score of 5 for Super, Regular Plus and HNI

4 Factor plot between Churn, Cashback and Complain Ly

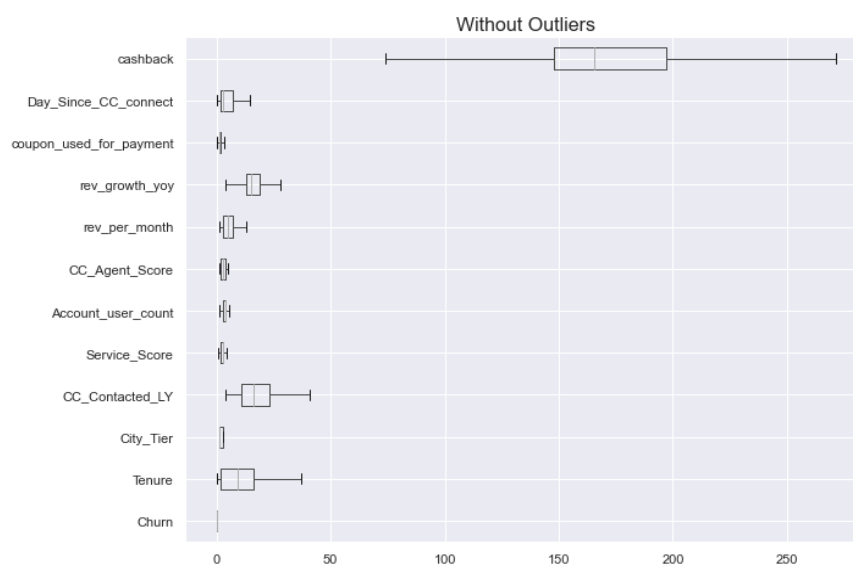


Median cashback for account users who have raised complaint and churned is lesser as compared to those who have not raised complaint and not churned

Data Preparation

1 Outlier Treatment:

We use Interquartile range to set lower and upper range and thus treat outliers using boxplot



Insight: We observe that all the outliers are removed

Business Insights from EDA:

The given dataset is an extract of Customer Account Churn Details of an E Commerce Company. The Dataset contains information like Account ID, Churning Yes or No, Payment Type, Account Segment Type, Service Score, Customer Agent Score, Tenure of the User Account, Complaints raised by user in last 12 months, Revenue per Month, Revenue Growth YOY, Days since last customer care connect, Gender, Marital Status, Cashback and coupons used for payment. After running through pre-processing and cleansing of data from checking for duplicates to missing values, removing bad data, changing the variable type, data visualization to checking for outliers...we have tried to prepare data before it can be split to train and test sets for making the predictive model.

1. All the variables in the given dataset are not important for preparing the model. Since we are looking at features which are directly impacting the Account Churn we may ignore few variables. In the first step post reading the dataset and getting the shape we drop the AccountID column
2. Also we see that the proportion of 0 and 1 for Churn variable is not balanced, Hence we deduce that the dataset is imbalanced and will later need treatment with VIF or Smote to balance the dataset.
3. Variables like Revenue per month and Revenue Growth YOY though important to business from financial point of view does not hold much relevance here as it does not affect the customer account churn
4. The highest correlation is between Churn and Account User Count and Churn and Customer Care Agent Score. The median no of users per account for accounts that churned is at 4 which is higher than account user count that have not churned. This means that accounts that have churned have higher number of users and hence caused more loss.
5. Maximum Churn is by users of Account Segment Regular Plus at 10%. Other Category have very minimal percentage from 2% to 3% churning
6. In Payment type users with Debit Card usage at maximum and also churning at highest rate of 6.4%
7. Though Tenure has lesser effect on Account Churning but it does have impact and the distribution is quite significant to study as most of users who have churned have tenure ranging between 0 and 20 months with median value between 0 and 1 month.

The End

